SDG HACKATHON PITCH

FuturePath AI

By NovaMind



Problem Statement

How might we build an AI-powered solution that helps students and fresh graduates that struggles to understand which technology career align with their skills, what competencies they lack of and how to prepare for changing workforce demands.

Target user

- **Undergraduate Students:** Students selecting majors, electives, or internships who need early foresight into high-demand future skills.
- **Graduating Seniors & Fresh Graduates:** Job seekers entering the workforce who need to evaluate their readiness for entry-level roles, identify skill deficiencies, and pivot toward future-proof career specializations.
- **Early-Career Switchers:** Recent graduates looking to transition into higher-growth, AI-adjacent sectors.



Why current solutions are not enough

Chatbot-Only Advice: It only provides career advice, but our platform provides structured weighted matching, transparent skill gaps and documented data source

One-Time Skills Assessments: Static skill quizzes fail because AI progress is continuous and unpredictable.

Lack of SDG & Impact Connection: Traditional career platforms ignore how workforce roles connect to broader sustainability and ESG goals.

Our solution

- **Dated job-market snapshot:** Data capturing the exact state of hiring trends and industry demands at a specific point in time.
- **Explainable weighted career matching:** A system that prioritizes your key criteria (like skills or experience) to suggest career paths and clearly explains the logic behind its recommendations.

- **Skill-gap analysis and reskilling plans:** Identifying the missing skills required for a target role and providing a structured roadmap to learn them.
- **Transparent sources and responsible guidance:** Disclosing the origins of the career data to ensure the advice provided is realistic, verifiable, and unbiased.

How it works

Select Skills: Begin by inputting or selecting your current abilities and competencies in the system.

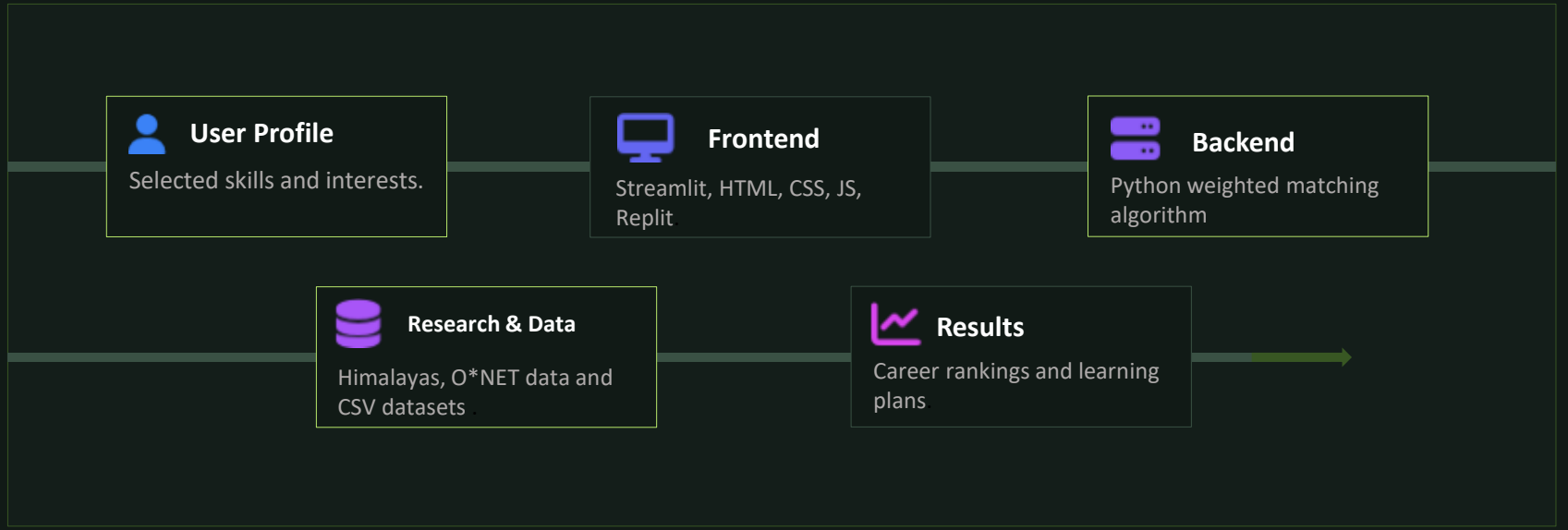
Compare Careers: The platform uses your selected skills to suggest potential job roles, allowing you to evaluate and weigh different career paths against your strengths.

Identify Gaps: System compares existing skill set against the specific requirements of a target career to highlight exactly what knowledge or abilities you are missing

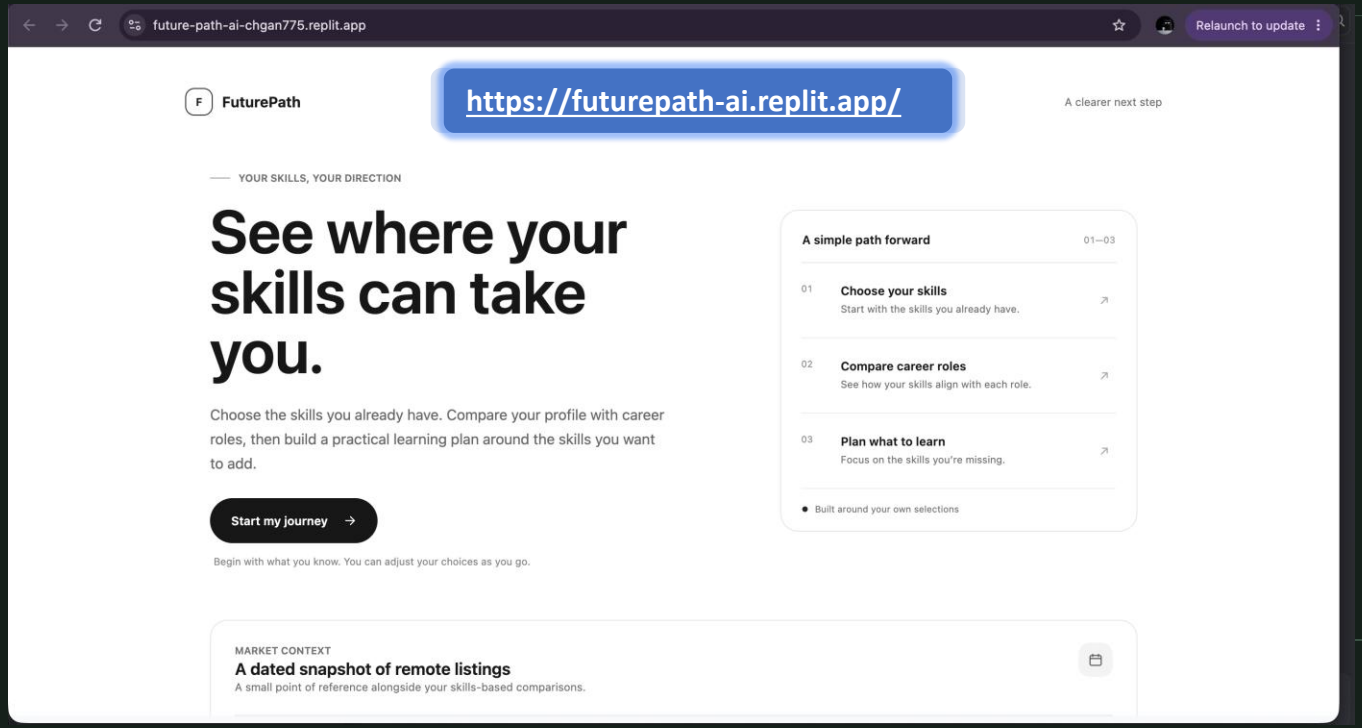
Create Learning Plan: Based on those identified gaps, the platform generates a structured roadmap or suggests specific training to help you acquire the missing skills.

Explore Market Snapshot: You review current data on hiring trends, industry demand, and job availability to validate that your chosen career path is viable in the present economy.

Technical architecture



Demo / prototype



Our Contribution to SDG impact

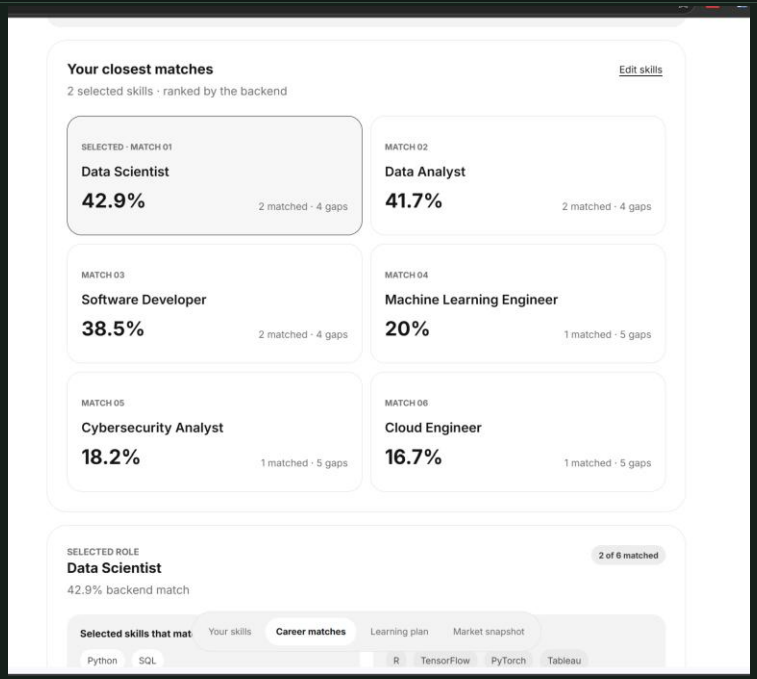
- **SDG 4 (Quality Education) - Target 4.4: Substantially increasing the number of youth and adults with relevant technical and vocational skills for employment and entrepreneurship.**

Our platform contributes by utilizing the "Identify Gaps" and "Create Learning Plan" workflow. By analyzing exactly what competencies a user is missing for a target role, the solution provides structured, actionable reskilling roadmaps, ensuring individuals acquire the precise, up-to-date skills actually demanded by employers.

- **SDG 8 (Decent Work & Economic Growth) - Target 8.5: Achieving full and productive employment and decent work for all women and men.**

Our solution contributes through "Explainable weighted career matching" and "Explore Market Snapshot" features. By transparently matching an individual's existing strengths with verified, real-time hiring trends and market demands, the platform minimizes skill mismatch in the labour market and empowers users to secure sustainable, productive roles that fit current economic realities.

Prototype Outcomes



Future improvements

Predictive Engine
Transition to live predictive market data.

Resume Parsing
Semantic CV profile analysis integration.

Model Upgrade
Advanced AI-powered career recommendations.

Dataset Expansion
Broader dataset integration and user validation.